

Lecture 05

Data-fitting, Evolutionary Optimization Models

Review on Lecture 03 & 04

- ❖ Engineering Economics:
Time Streams of Economic Benefits & Costs
- ❖ Solving Nonlinear Optimization Models with Calculus
 - ◆ Solution Using Calculus
 - ◆ Solution Using Hill Climbing
 - ◆ Solution Using the Lagrange Multiplier
- ❖ Concepts of Dynamic Programming Models (DP)
- ❖ Examples of Problem Solving with DP Models
- ❖ Concepts of Linear Programming Models (LP)
- ❖ Real-world Applications of Traditional Optimization Models

Curses of Stochastic Dynamic Programming

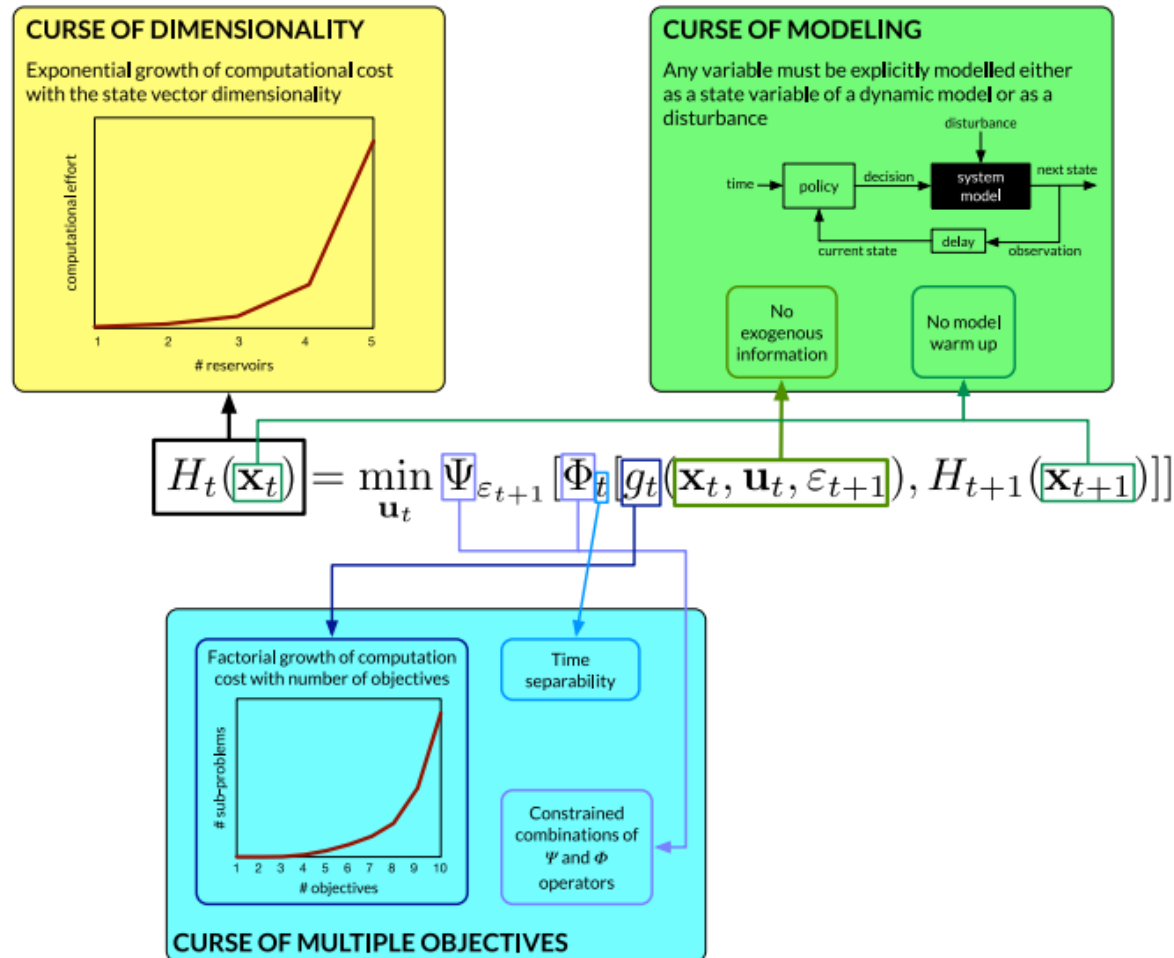


Figure 3. Schematization of the three curses of SDP with respect to the Bellman equation.

Before Lecture 05

❖ Acronyms (약어)/Keywords in Lecture 05

- ◆ EA: Evolutionary Algorithms (진화 알고리즘)
- ◆ GA: Genetic Algorithms (유전 알고리즘)
- ◆ GP/EP: Genetic Programming/Evolutionary Programming
(유전 프로그래밍/진화 프로그래밍)
- ◆ ANN: Artificial Neural Networks (인공 신경망)
- ◆ DE: Differential Evolutions (차분 진화 알고리즘)

❖ 향후 주요 일정

- ◆ Datacamp #2 due Apr-19
- ◆ Assignment #2 due Apr-26 (23:59까지 메일로 제출)
- ◆ 보령댐 견학 일정: 4/25 월 (확정, 공문 공유 완료) (No class on Apr-26 Tue)
- ◆ Term Project A due May-03
(PPT Presentation, prepare one color copy for the instructor)
- ◆ Quiz #2 on May-03

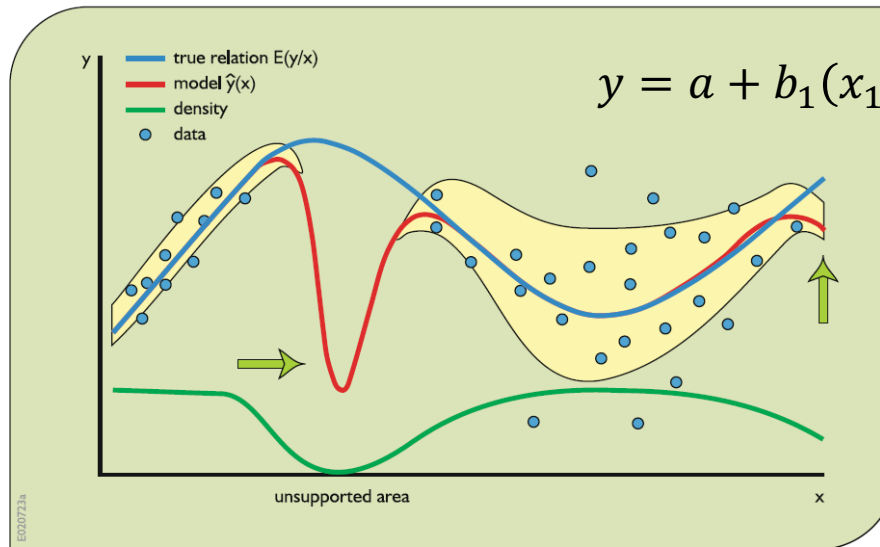
Today's Topic: Data-fitting, Evolutionary Optimization Models

- ❖ EA: Evolutionary Algorithms (진화 알고리즘)
- ❖ GA: Genetic Algorithms (유전 알고리즘)
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- ❖ ANN: Artificial Neural Networks (인공 신경망)
- ❖ DE: Differential Evolutions (차분 진화 알고리즘)

- ❖ EMODPS: Multi-objective Direct Policy Search
(진화적 다목적 직접 정책 검색법)

Examples of Data-driven Models: Regression

- ❖ Data-fitting or Statistical Models (ex. Regression)
 - ◆ Converts observed inputs (rainfall, inflow, etc.) to observed outputs (runoff, release, etc.) using mathematical equations
 - ◆ Depend on observed inputs and observed outputs for the estimation of the values of their parameters (a, b, c, etc.)



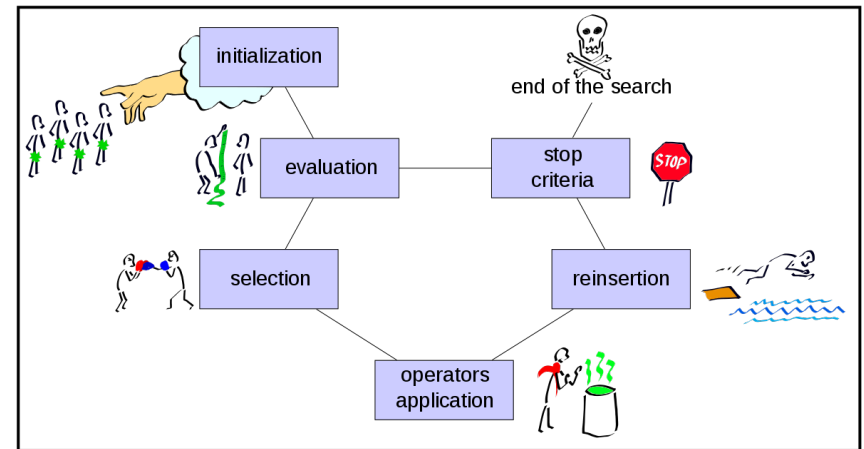
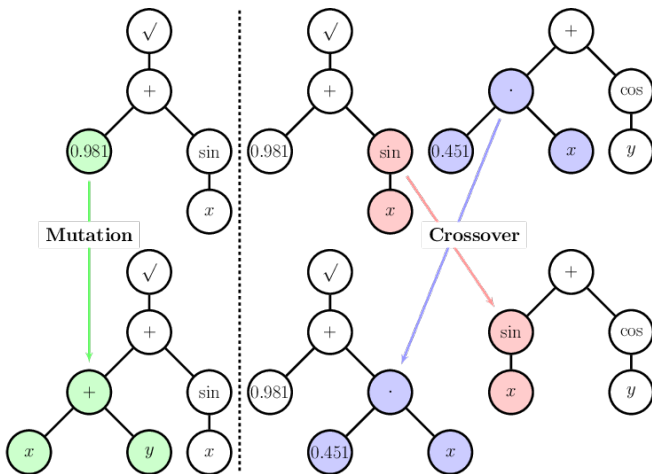
관측값 (x,y)으로부터
그에 맞는 매개변수값 (a,b)를
찾는 방법

Fig. 5.1 Data-fitting models are able to estimate relatively accurately within their calibrated ranges, but not outside those ranges. The bottom curve represents the

relative density of data used in model calibration. The arrows point to where the model does not predict well

Examples of Data-driven Models

- ❖ Many of them based on biological principles & concepts
- ❖ Some examples...
 - ◆ Evolutionary Algorithms (EA)
 - ◆ Genetic/evolutionary Programming (GP/EP)
 - ◆ Evolutionary Strategy (ES), ANN, etc.

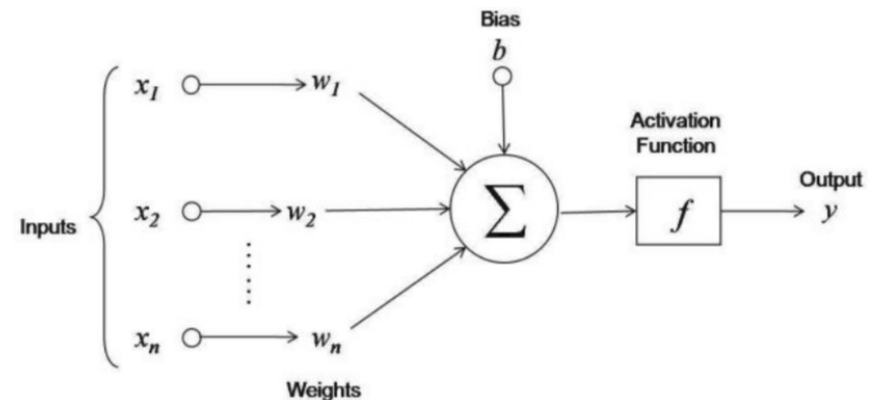
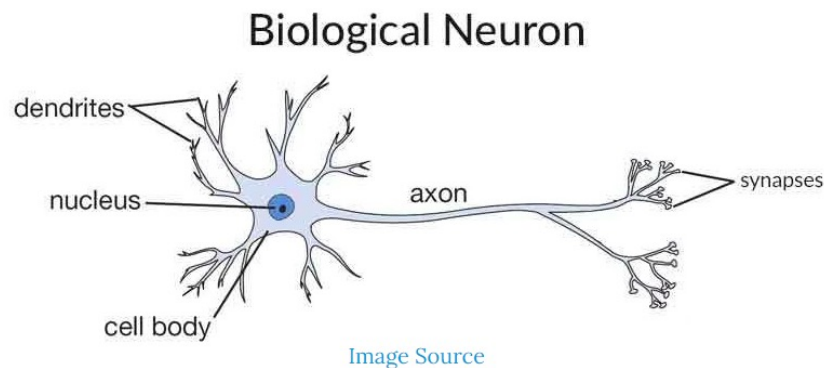


General schema of an Evolutionary Algorithm (EA)

Artificial Neural Network (ANN) Models

❖ Concepts behind the ANN Models

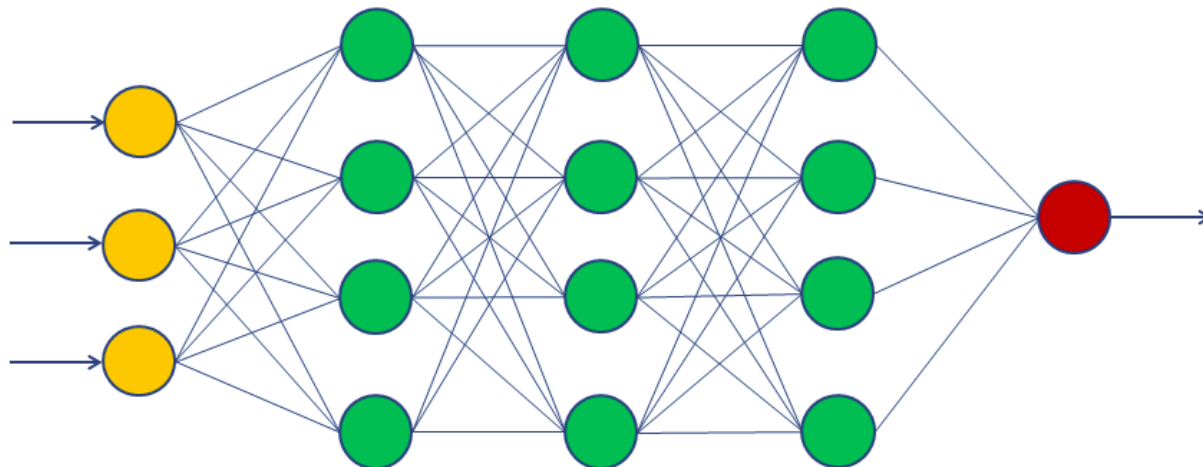
- ◆ Information processing model inspired by the biological neuron system
- ◆ Distributed, adaptive, generally nonlinear networks built from many different processing elements
- ◆ Just a more complex type of regression or statistical (black-box) model



Artificial Neural Network (ANN) Models

❖ Components of ANN models

- ◆ Input layer: nodes that receive an input from the external environment (send their weighted values to hidden layer nodes)
- ◆ Hidden layer: nodes that typically receive the transferred weighted inputs from the input/previous hidden layer
- ◆ Output layer: nodes that receive the hidden layer output and send it to the user



Artificial Neural Network (ANN) Models

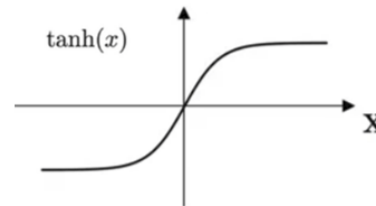
❖ Types of ANN Models

- ◆ Feedforward ANN- NOT recursive, signals travel in ONLY ONE direction towards the output layer
- ◆ Feedback ANN (=Recurrent Neural Networks, RNN)- recursive, past inputs can be fed back into with current inputs

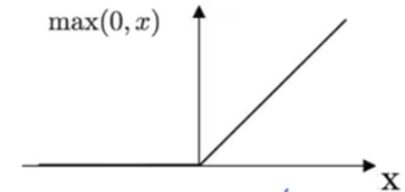
❖ Activation Functions

- ◆ Identity function
- ◆ Binary Step Function
- ◆ Sigmoid Function
- ◆ Ramp Function
- ◆ ReLU (Rectified Linear unit)

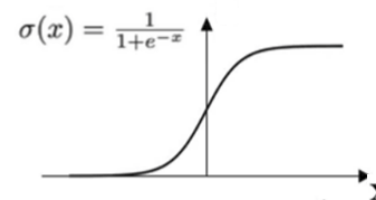
Hyper Tangent Function



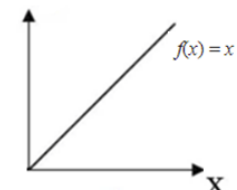
ReLU Function *



Sigmoid Function



Identity Function



Artificial Neural Network (ANN) Models

❖ Procedures for preparing/training an ANN (Smith 1993)

1. Design a network.
2. Divide the dataset into training, validation, and testing subsets.
3. Train the network on the training data set.
4. Periodically stop the training and measure the error on the validation data set.
5. Save the weights of the network.
6. Repeat Steps 2, 3, 4 until the error on the validation data sets starts increasing. This is the moment where the overfitting has started.
7. Go back to the weights that produced the lowest error on the validation data set, and use these weights for the trained ANN.
8. Test the trained ANN using the testing data set. If it shows good performance, use it. If not, redesign the network and repeat the entire procedure from Step 3.

ANN Example (Binary)

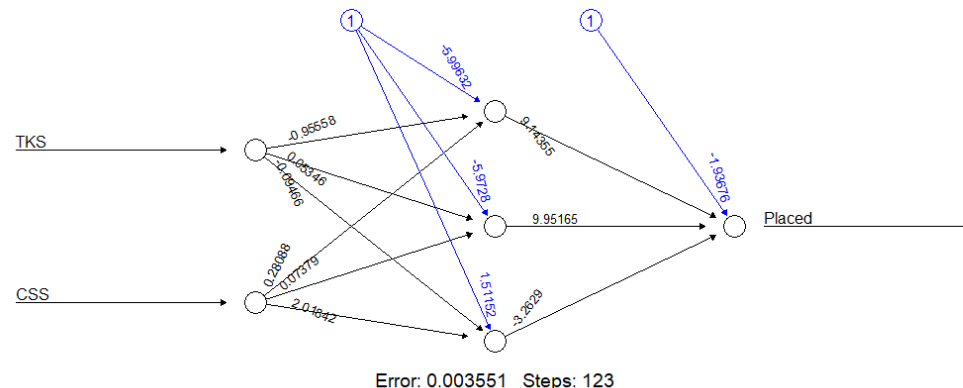
Create training dataset

❖ Training Dataset

Technical Knowledge Score	Communication Skills Score	Student Placed
20	90	Placed
10	20	Not Placed
30	40	Not Placed
20	50	Not Placed
80	50	Placed
30	80	Placed

❖ Train the model with given data (3 hidden layers with logistic activation function)

❖ Result



ANN Example (Binary)

❖ Test the model with Test dataset

Technical Knowledge Score	Communication Skills Score	Student Placed
30	85	?
40	50	?
85	40	?

```
> Predict$net.result
      [,1]
[1,] 0.9695585
[2,] 0.3581687
[3,] 0.9512383
> # Converting probabilities into binary classes setting threshold level 0.5
> prob <- Predict$net.result
> pred <- ifelse(prob>0.5, 1, 0)
> pred
      [,1]
[1,] 1
[2,] 0
[3,] 1
```

ANN Example (Binary)

```
#https://www.datacamp.com/community/tutorials/neural-network-models-r
#install.packages("neuralnet")
library(neuralnet)

set.seed(42)
# creating training data set
TKS=c(20,10,30,20,80,30)
CSS=c(90,20,40,50,50,80)
Placed=c(1,0,0,0,1,1)
# Here, you will combine multiple columns or features into a single set of data
df=data.frame(TKS,CSS,Placed)

nn=neuralnet(Placed~TKS+CSS,data=df, hidden=3,act.fct = "logistic",
             linear.output = FALSE)

# plot neural network
plot(nn)

# creating test set
TKS=c(30,40,85)
CSS=c(85,50,40)
test=data.frame(TKS,CSS)

## Prediction using neural network
Predict=compute(nn,test)
Predict$net.result

# Converting probabilities into binary classes setting threshold level 0.5
prob <- Predict$net.result
pred <- ifelse(prob>0.5, 1, 0)
pred
```

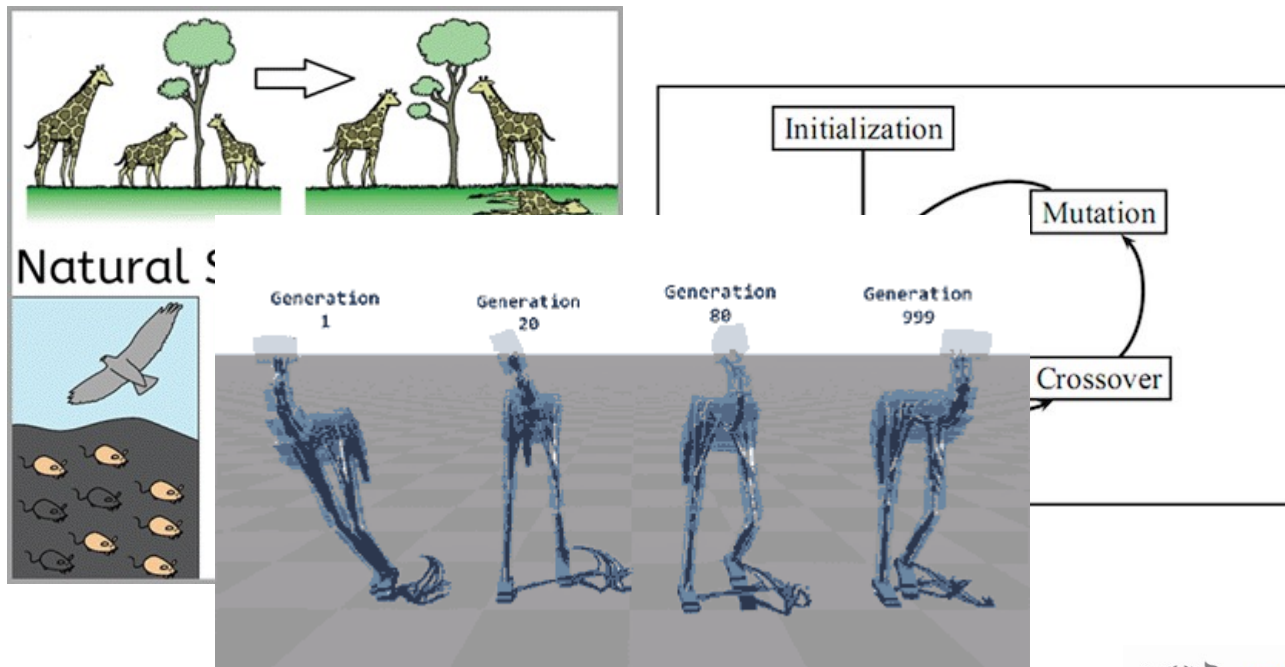
ANN Models can be used in:

- 1) Pattern Recognition, such as facial/fingerprint recognition, and object detection
- 2) Anomaly detection
- 3) Time Series Prediction, such as stock price and weather forecasting
- 4) Natural Language Processing

Evolutionary Algorithms (EA)

❖ Evolutionary Algorithms

- ◆ Randomized general-purpose search techniques used for finding the best values of the parameters or decision variables
 - Based on the mechanisms of natural selection (Goldberg 1989)



Evolutionary Algorithms (EA)

❖ Evolutionary Algorithms

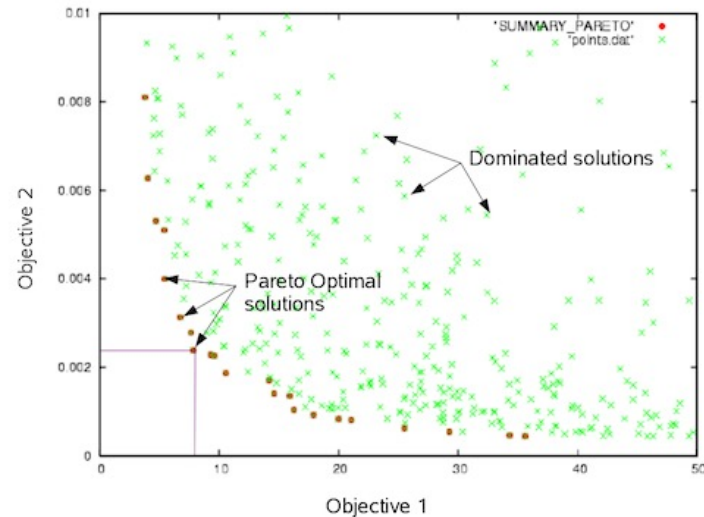
◆ Broad spectrum of heuristic approaches for simulating biological evolution in the search for improved “fitness”

- Fitness = the best values of decision variables and parameters based on an objective or fitness function

◆ Based on mutation, recombination, & selection

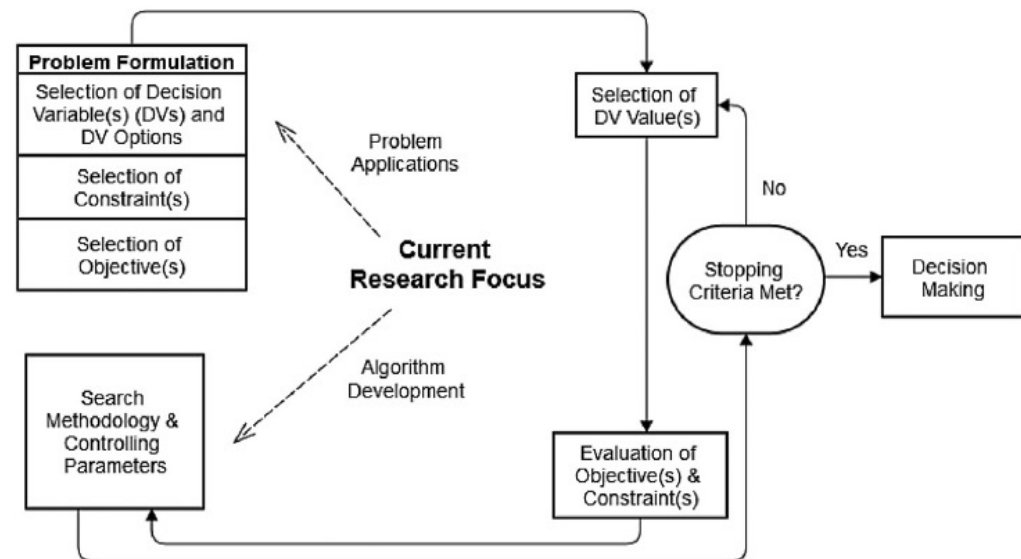
◆ Some examples

- GA, ES, EP, GP
- ACO, PSO, BFO, etc.



Evolutionary Algorithms (EA)

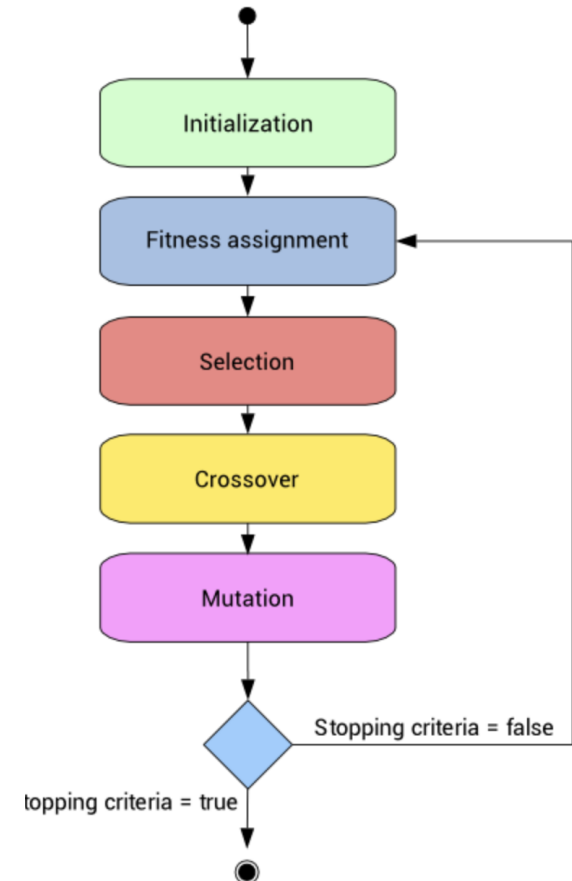
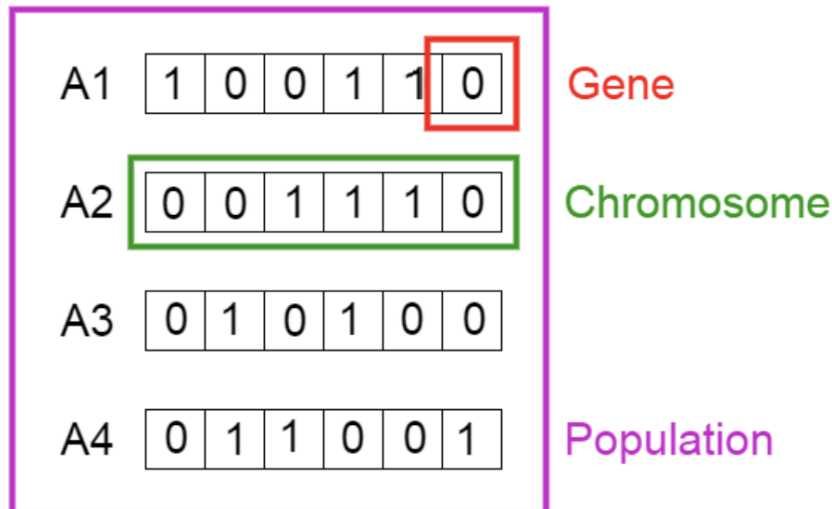
- ❖ Steps in the optimization process using EAs (Maier et al. 2014)
 - 1) Problem formulation (i.e. selection and definition of decision variables, objectives, and constraints).
 - 2) Selection of decision variable values.
 - 3) Evaluation of objectives and constraints for the selected decision variable values, which is generally done using one or more simulation models.
 - 4) Selection of an improved solution based on the feedback received from the simulation models.
 - 5) Repetition of steps 2-4 until a satisfactory solution has been satisfied.
 - 6) Passing the optimized solution to the decision making process.



Evolutionary Algorithms (EA)

❖ Elements of GA

- ◆ Population
- ◆ Chromosomes
- ◆ Genes



Evolutionary Algorithms (EA)

Population

	Locus (Position)									
	1	2	3	4	5	6	7	8	9	10
	1	0	1	1	1	1	0	0	1	0
	1	0	1	0	0	0	1	1	1	0
	0	0	1	1	1	1	1	0	0	0
	0	1	0	0	1	0	1	1	1	1
	0	1	1	0	0	1	0	0	1	0
	0	1	1	1	0	1	1	0	0	0
	1	0	0	0	0	1	1	1	1	1
	1	0	1	1	0	0	0	1	0	1
	0	0	1	1	0	1	0	1	1	0
	0	0	1	0	1	0	1	1	0	0

Allele (Value) = 0 AI

Fig. 1.8. Generalized E

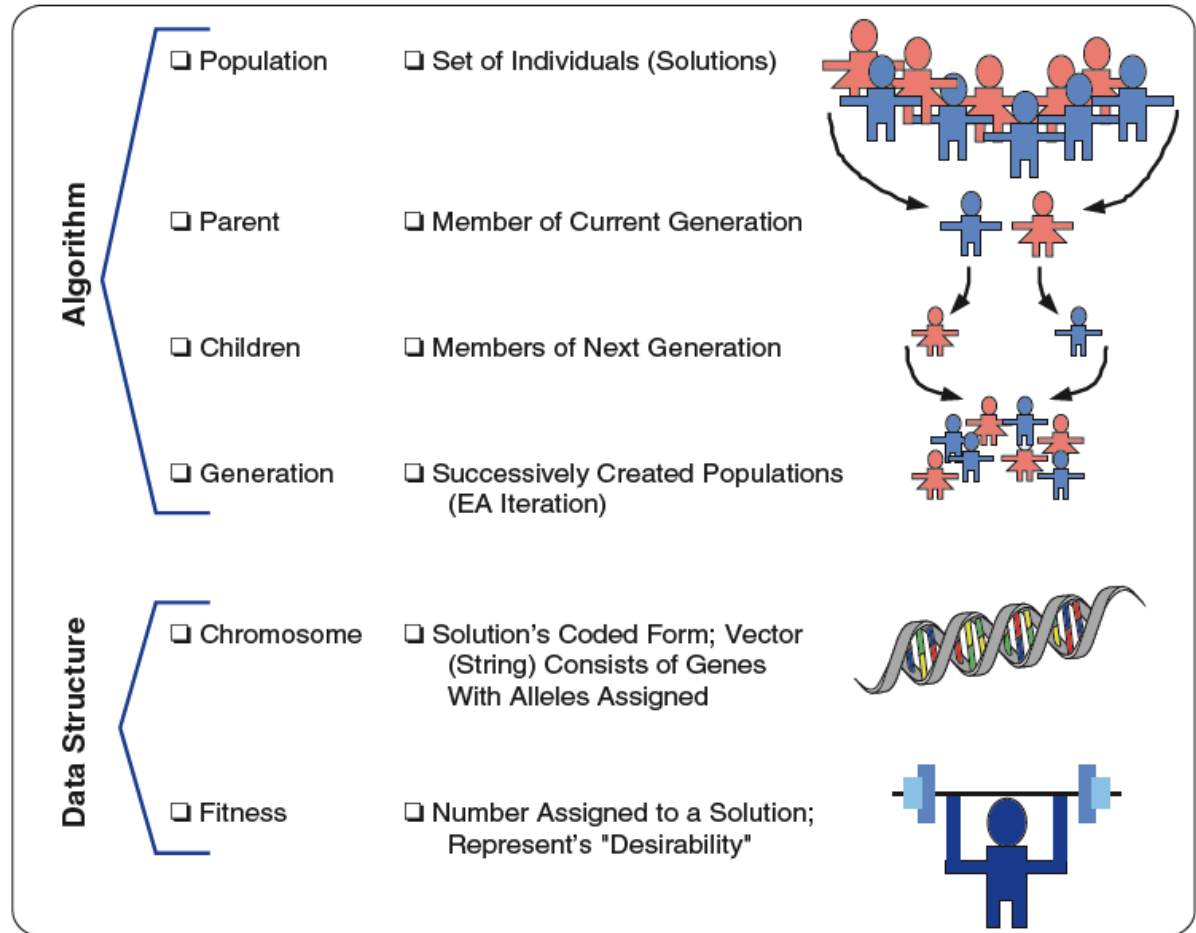


Fig. 1.9. Key EA Components

EA Examples

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1														
2		x1	x2		Obj									
3		0.808239	12.37206		13733.07									
4														
5	C1			-0.06423	<=	0								
6	C2			0.000418	<=	0								
7	C3			0.808239	<=	1								
8	C4			12.37206	<=	13								
9														
10														
11														
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해 찾기 매개 변수

목표 설정(T)

대상: ☐ 최대값(M) ☒ 최소(N) ☐ 지정값(V)

변수 셀 변경(B)

제한 조건에 종속(U)

\$D\$5:\$D\$8 <= \$F\$5:\$F\$8
 \$D\$7 >= 0
 \$D\$8 >= 0

☒ 제한되지 않는 변수를 음이 아닌 수로 설정(K)

해법 선택(E)

해법

완전한 비선형으로 구성된 해 찾기 문제에 대해서는 GRG Nonlinear 엔진을 선택합니다.
 선형 문제에 대해서는 LP Simplex 엔진을 선택하고 완전하지 않은 비선형으로 구성된 해
 찾기 문제에 대해서는 Evolutionary 엔진을 선택합니다.

도움말(H)

Genetic Programming (GP)

❖ Genetic Programming (GP)

- ◆ Induces a population of computer programs or models that improve automatically as they experience the data on which they are trained

❖ GP vs GA? 가장 큰 차이점은?

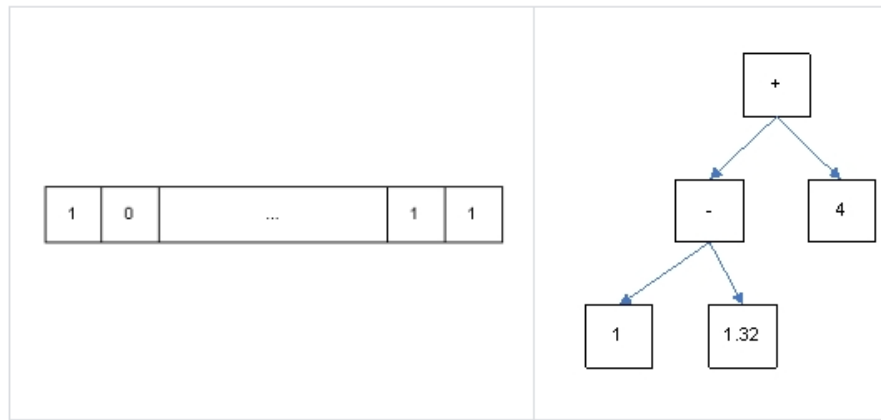
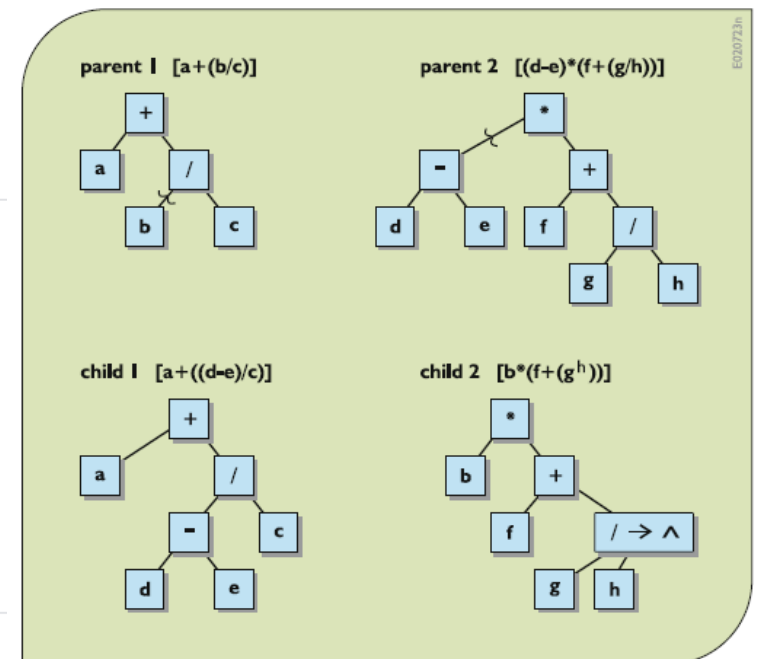


그림1. GA 와 GP 의 표현의 차이점



Curses of Stochastic Dynamic Programming

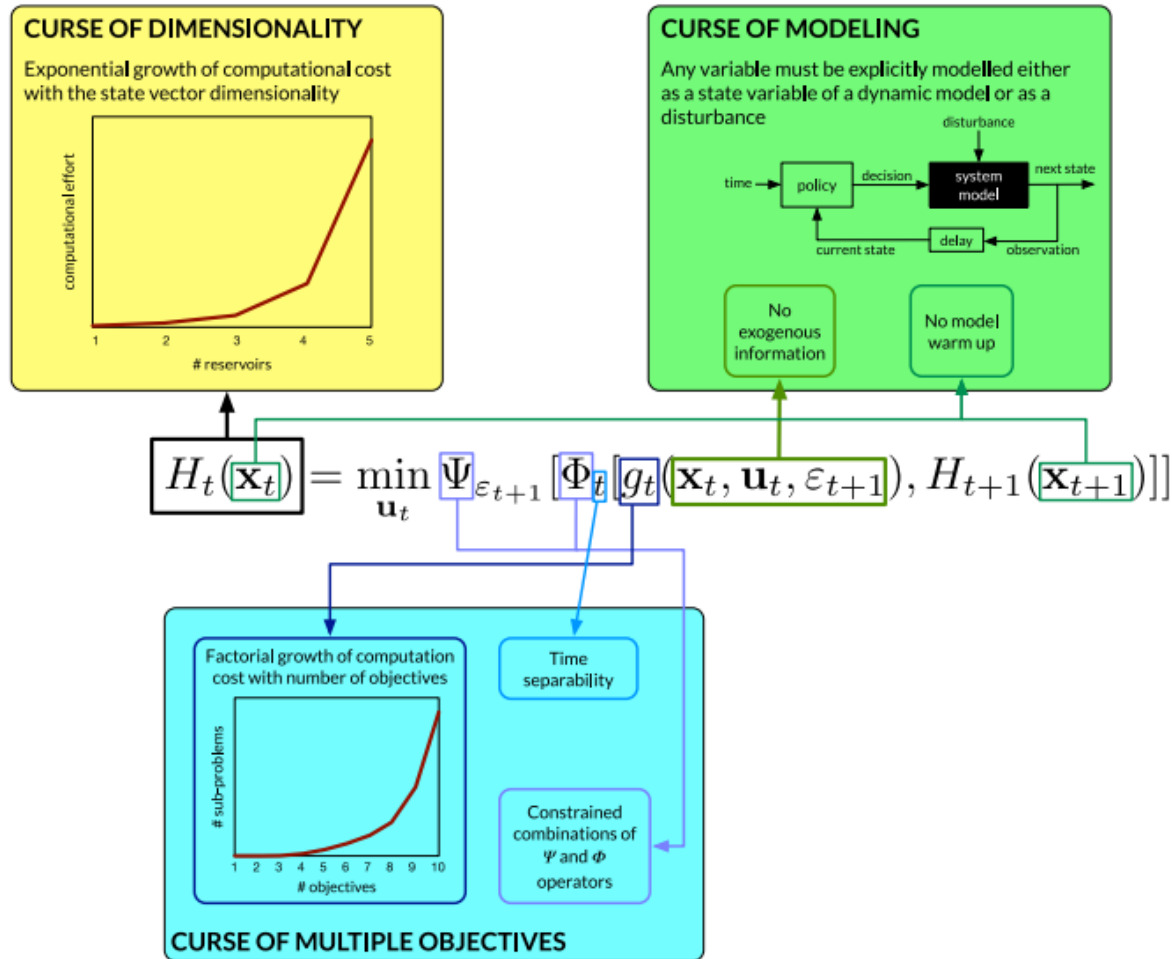
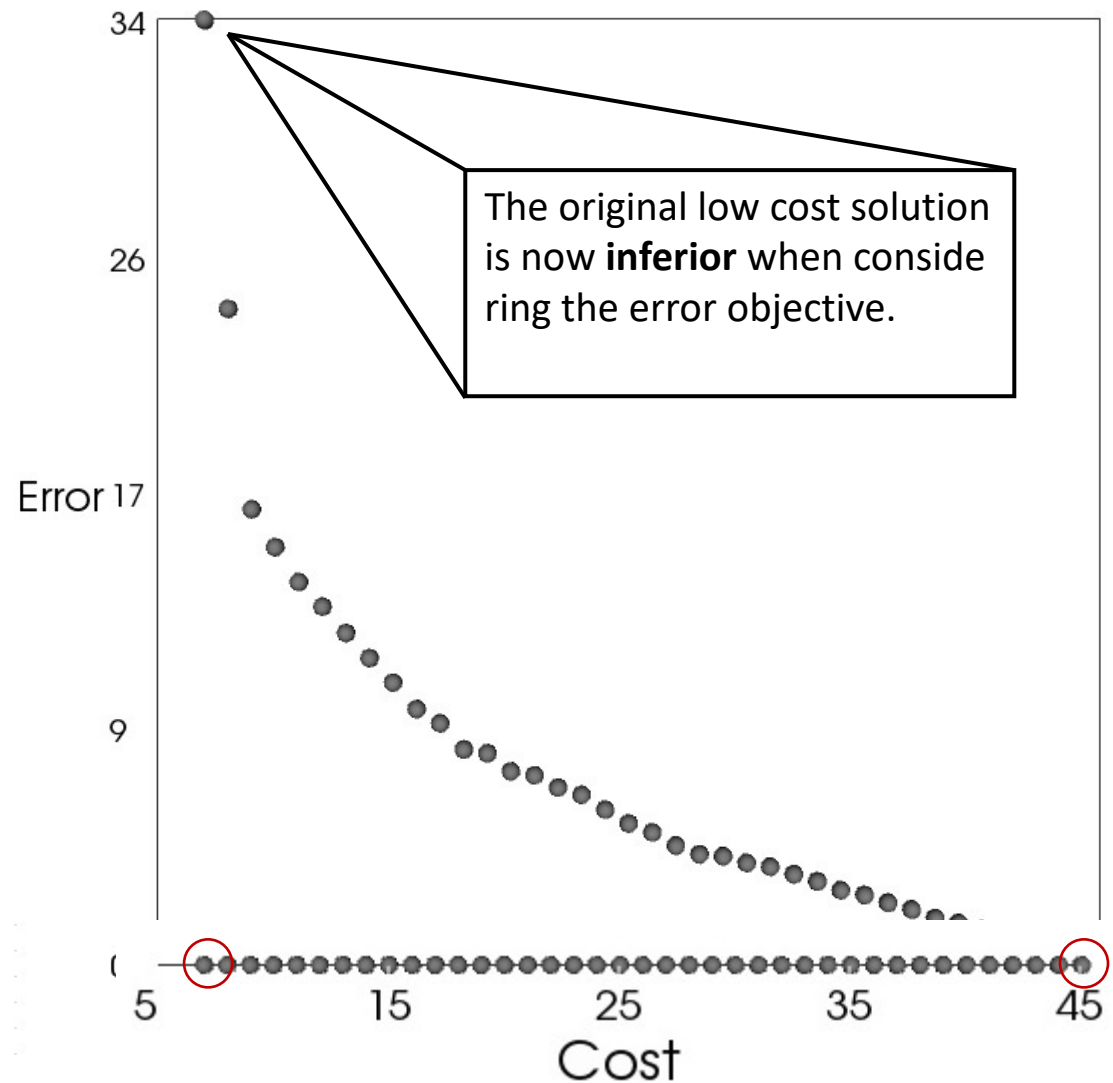


Figure 3. Schematization of the three curses of SDP with respect to the Bellman equation.

The Benefits of Many Objectives

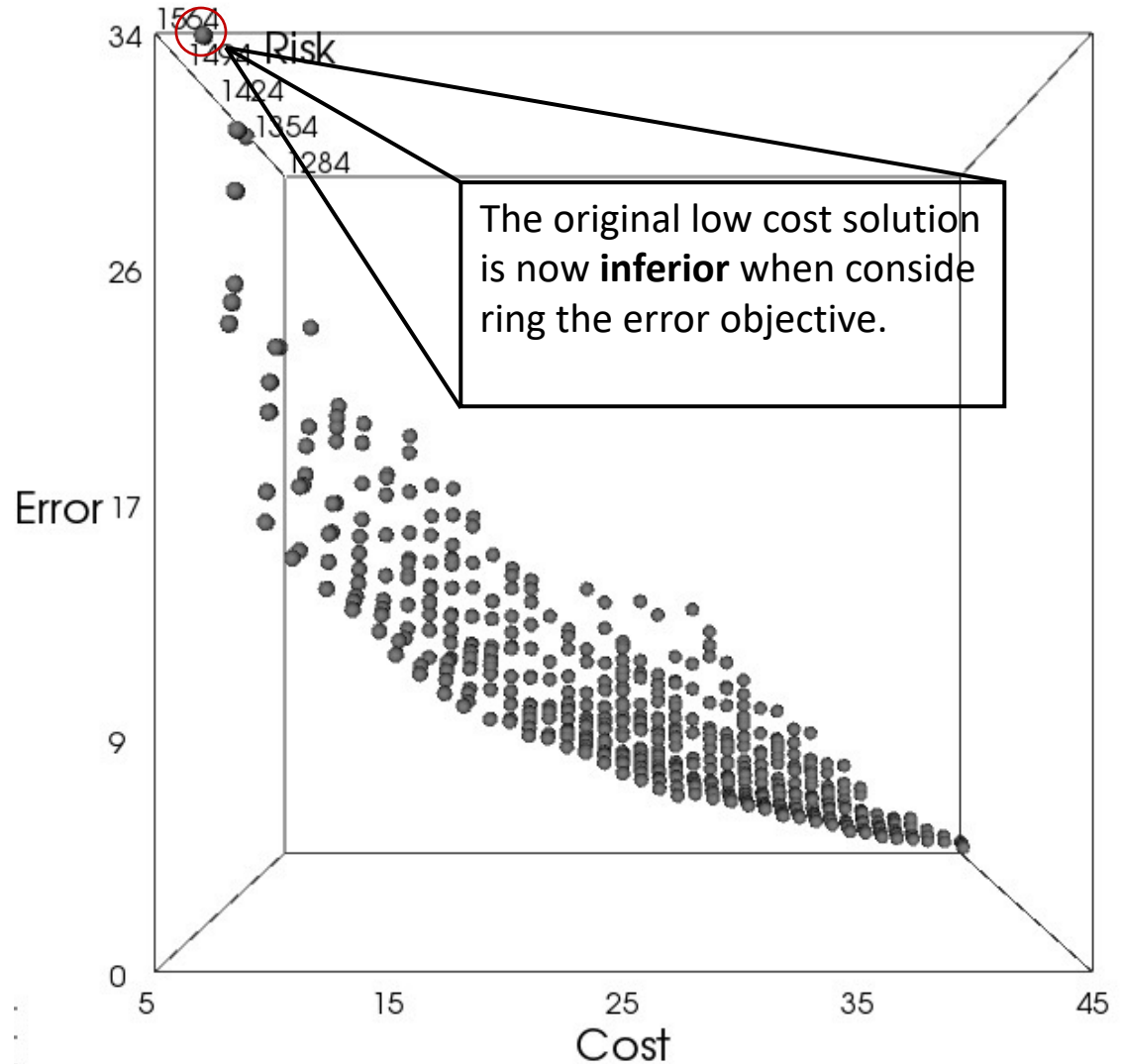
- Single Objective Design Problem...
- Two Objective Design Problem...
- Many-Objective Design Problem...

- Two and three objective subsets
- Difficult to specify manually



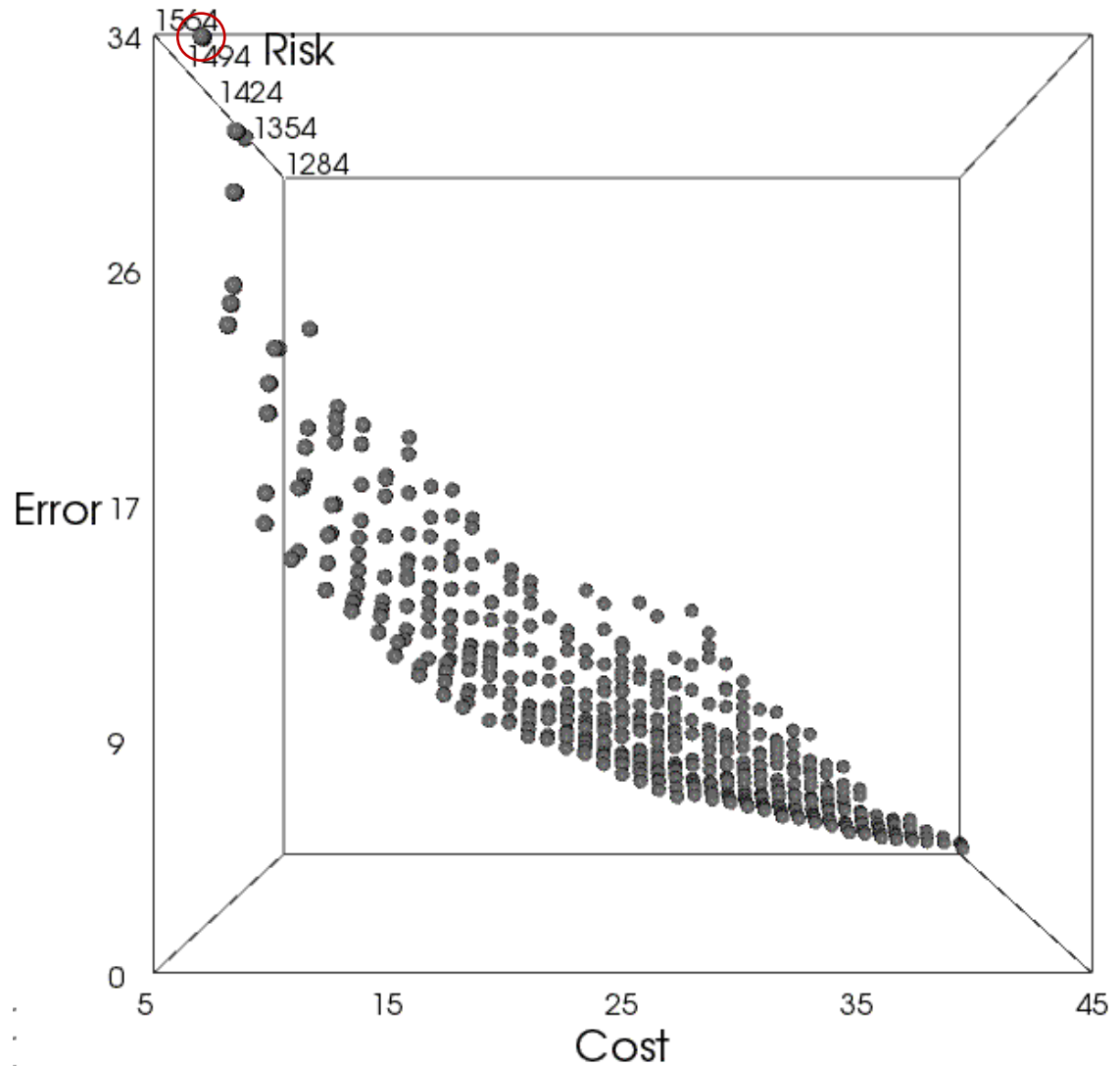
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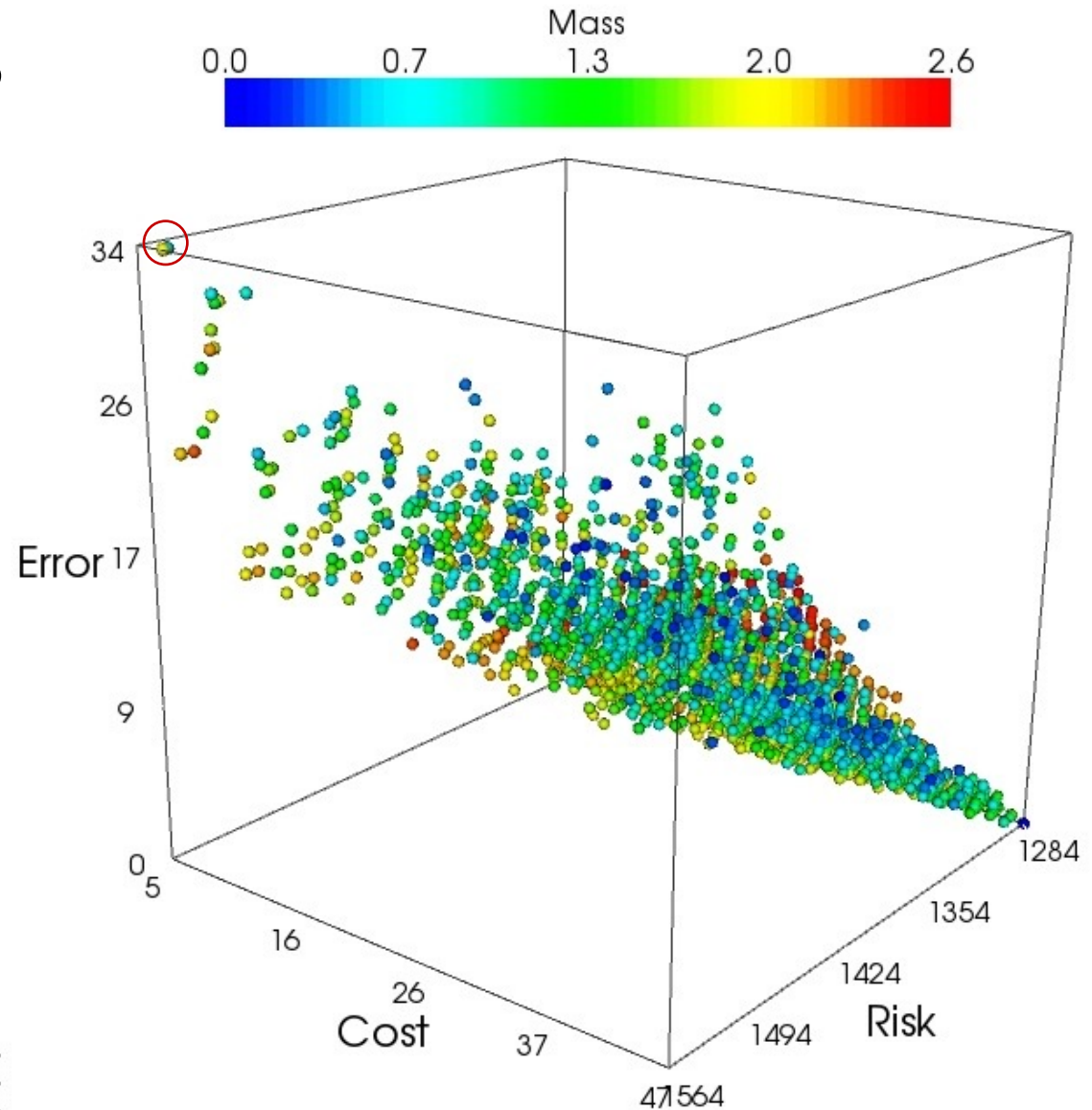
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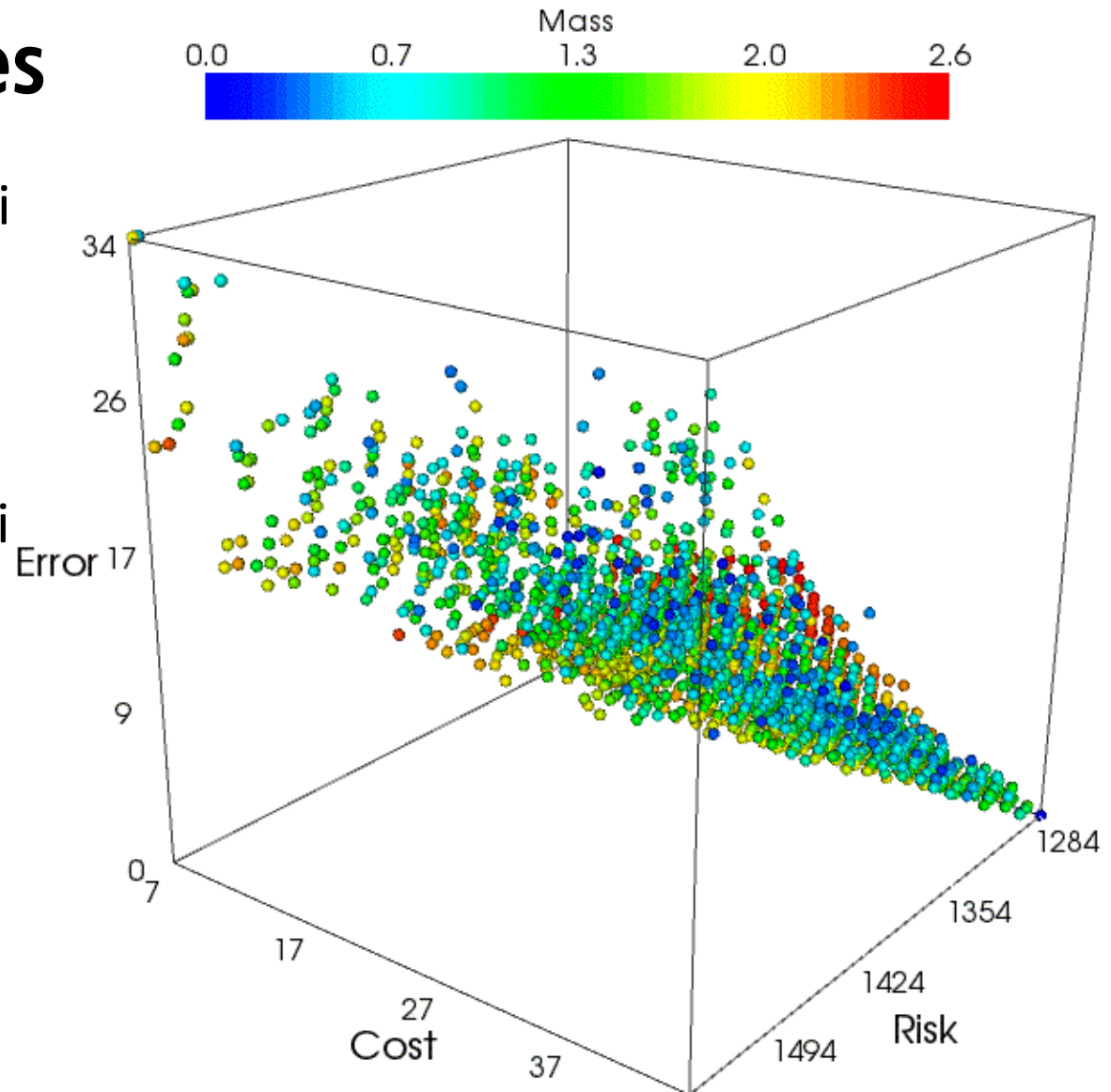
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The Benefits of Many Objectives

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Examples of models based on EAs

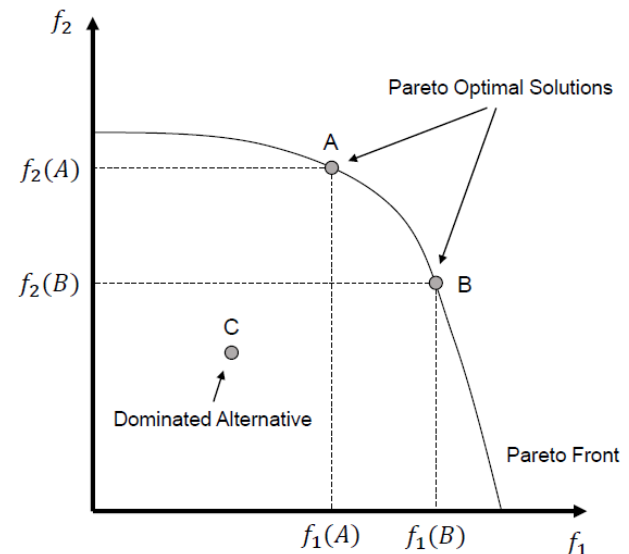
❖ Multi-objective Evolutionary Algorithms (MOEAs) (Schaffer 1985)

- ◆ Targets on the explicit representation of the tradeoff between multiple conflicting objectives in multi-objective problem (Coello Coello et al. 2007)

- ◆ Based on the concept of Pareto Optimality

(Ex) Pareto Optimal solution X dominates inferior solution Y in a maximization problem with i objectives: $(f_1, f_2, \dots, f_i)$

- 1) For all j , $F_j(X) \geq F_j(Y)$
- 2) For at least one j among i objectives, $F_j(X) > F_j(Y)$



Example: EMODPS

❖ Evolutionary Multi-Objective Direct Policy Search (EMODPS)

◆ Direct Policy Search (DPS) (Rosenstein and Barto, 2001)

- A type of parameterization-simulation-optimization (PSO) approach in which the decision variable directly operates in the policy space

◆ Nonlinear approximating networks: Radial Basis Function (RBFs)

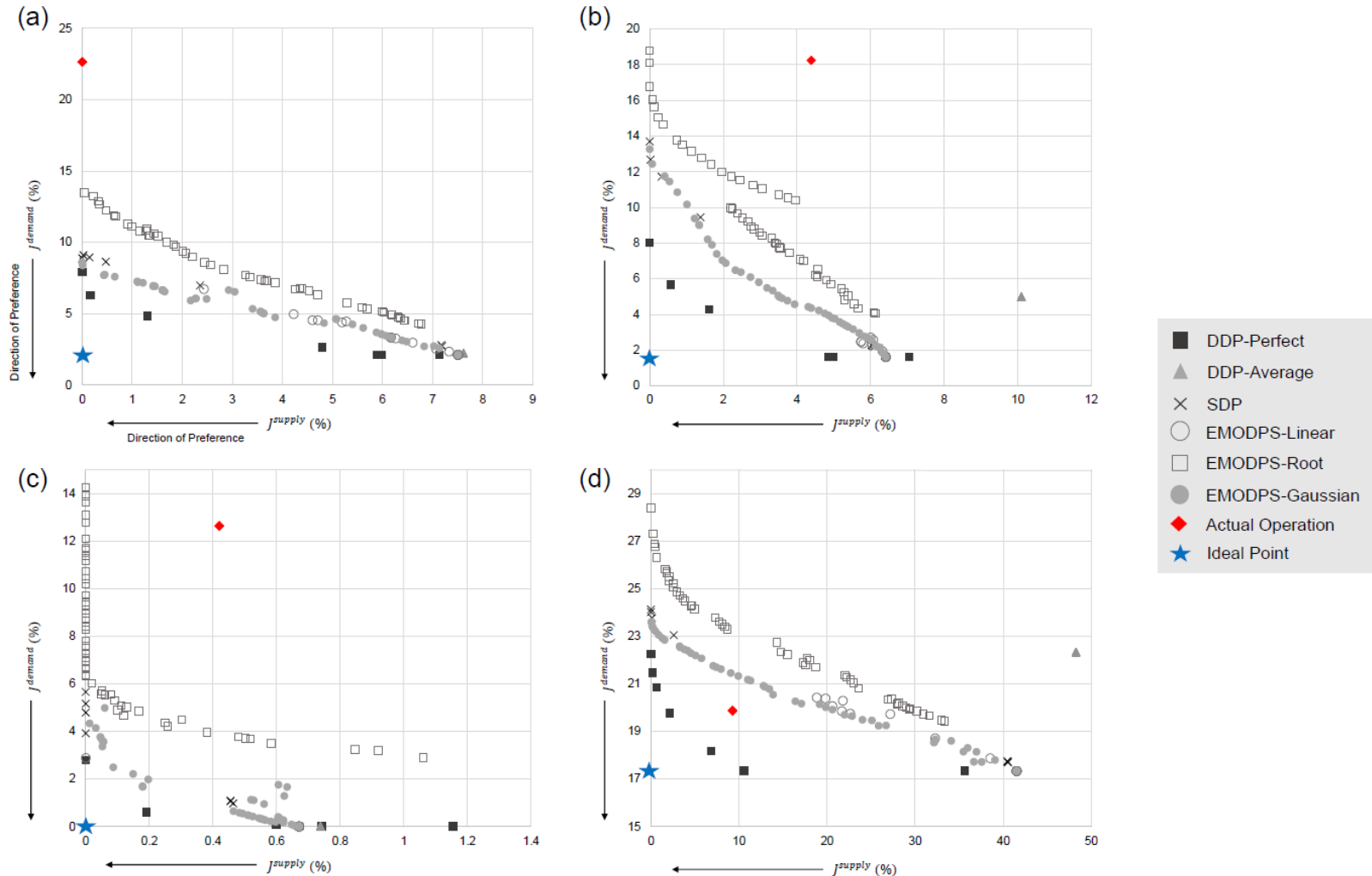
- Approximating function that determines the value of the release decisions

◆ MOEAs during the solution searching process

- Iterative search algorithm that evolve a Pareto-approximate set of solutions in a multi-objective problem

$$R_t = \min \left(\max \left(\sum_{i=1}^j \left(w_i \times \left| D_t - \frac{S_t - d_i}{r_i} \right|^{0.5} \right), R_{\max} \right), R_{\min} \right)$$

Example: EMODPS



Conclusion of Lecture 05

❖ Key takeaways / Thoughts?

- ◆ How does the evolutionary optimizations models differ with the conventional optimization models (LP, DP)? What are the key assumptions made in each type of model?
- ◆ Possible limitations of data-driven optimization models?
 - “Black Box”, Outliers, anything else?

❖ Possible Future Topics?

- ◆ Uncertainty!
- ◆ Chapter 6: An Introduction to Probability, Statistics, and Uncertainty

Future Schedule

- ❖ 다음 주 강의 (2022-04-19 화요일 17:00-19:50)
 - ◆ Training with Evolutionary Optimization Models (Laptops)
 - INSTALL R & RSTUDIO in your laptops
 - ◆ Future Schedule Reminder
 - Datacamp #2 due Apr-19
 - Assignment #2 due Apr-26 (23:59까지 메일로 제출)
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